

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

COMPUTER GRAPHICS

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SCOPE

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2D SCALING

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2D SCALING OVERVIEW?

A 2D scaling is applied to an object by zooming or scaling(change size) it along a **path** in XY plane, to **scale** the object a scaling factors (S_x, S_y) is **multiplied** by (x,y) coordinations values of each vertex of the polygon.

1- SCALING BY

Fixed Point (x,y)

2- SCALING BY

Pivot point (Arbitrary Point)

When scaling factor $S(S_x, S_y) > 1$ then scale up, and When $S(S_x \text{ or } S_y) < 1$ then scale down, and when $S(S_x \text{ or } S_y) = 1$ then no scaling is done.

When $S_x = S_y$ no change the shape of the object(Uniform Scaling).

When $S_x \neq S_y$ no change the shape of the object(Uniform Scaling).



2D SCALING

Formula?

□ Scale about the origin(Fixed) formulas

$$X' = x \cdot S_x$$
$$Y' = y \cdot S_y$$

Let us assume that the original **coordinates** are **(x, y)**, the scaling factors are **(S_x, S_y)**, and the produced **coordinates** are **(x', y')**. This can be **mathematically** represented as shown left

➤ Formula as a matrix

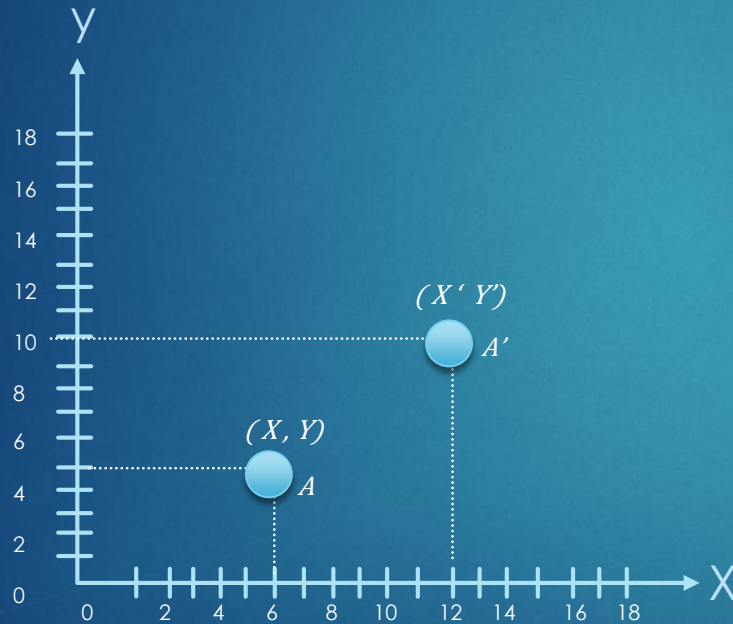
$$P' = P * S$$

$$\begin{bmatrix} X' & Y' \end{bmatrix} = \begin{bmatrix} X & Y \end{bmatrix} * \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$



EXAMPLE OF SCALING

✓ Scale a point with a coordinate of $A (6,5)$ into $(2,2)$



$$[X' \quad Y'] = [X \quad Y] * \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

$$[6 \quad 5] * \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$[12 \quad 0, \quad 0 \quad 10]$$

$$[x' \quad y'] = [12, 10]$$

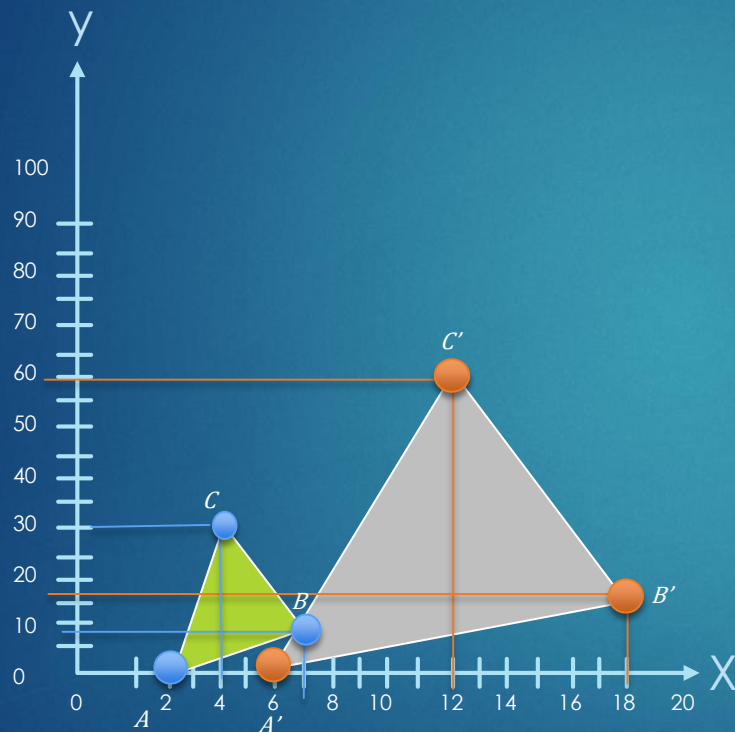
$$[12, 10]$$

$$P' = (12, 10)$$



EXAMPLE OF TRANSLATION

- ✓ A triangle with vertices **(2,0)**, **(7,8)**, and **(4,30)** been scaled **(3,2) units**. What are the new coordinates of this triangle after scaling .



$$X' = X \cdot S_x$$

$$Y' = Y \cdot S_y$$

$$A' = \begin{matrix} 2 & * & 3 & = & 6 \\ 0 & * & 2 & = & 0 \end{matrix}$$

$$A' = (6, 0)$$

$$B' = \begin{matrix} 6 & * & 3 & = & 18 \\ 8 & * & 2 & = & 16 \end{matrix}$$

$$B' = (18, 16)$$

$$C' = \begin{matrix} 4 & * & 3 & = & 12 \\ 30 & * & 2 & = & 60 \end{matrix}$$

$$C' = (12, 60)$$





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THE END

